

Regularisation

→ It is all about maintaining a balance between the penalizing addition of new features to the model and Reduction of loss function or we can say Bias and variance

→ It prevents overfitting by limiting the search space (for the loss function) by placing bounds or guard regions near origin. The value of loss function can't cross the bounds due to limitations placed by the bounds

L1 Norms (Lasso)

$$L_1 \text{ term} = |\beta_i| = |\beta_0| + |\beta_1| + |\beta_2| + \dots + |\beta_k|$$

It is sum of the absolute values of all the weight terms such as $\beta_0, \beta_1, \dots, \beta_k$.

→ In 2D space this can be viewed as weight matrix consisting of two terms $\beta = [\beta_1, \beta_2]$ or we can say $\beta_i = \beta_1 + \beta_2$

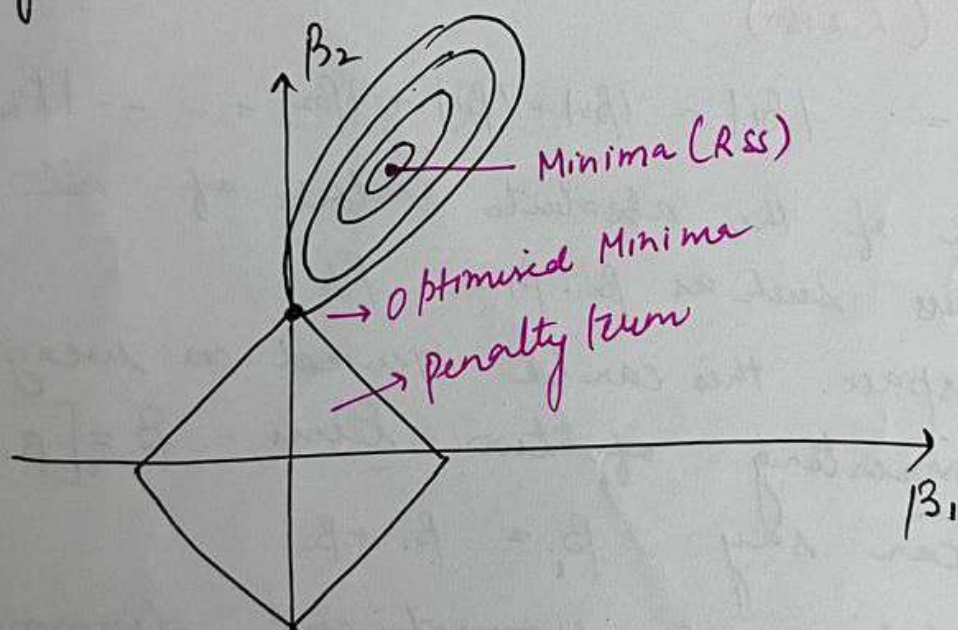
which defines a symmetrical diamond or rhombus around origin. This shape is similar to one obtained from equation of line $y = mx + c$

→ L1 norms defines a diamond shaped boundary around origin which restricts the loss function values from obtaining a value of 0, this prevents the model from overfitting. During optimization of loss function, the value of loss

of the loss function will not decrease further than the value obtained when elliptical contour touches/intersects the boundary defined by the norm

→ In case of L_1 Regularisation, the loss function has a greater chance of hitting the edges of diamond rather than the sides due to extreme shape of the boundary.

→ Hence L_1 regularisation results in some weights turning to 0.



[for this situation $\beta_1 = 0$]

So Lasso creates a sparse and light weight model with many weights value set to 0. This enables us to use L_1 regularisation as a feature selector as it discards some weight values.

norms

Ridge

2 term which is added to the end of the loss function is defined as:

$$|\beta_i|^2 = (\beta_0^2 + \beta_1^2 + \beta_2^2 + \dots + \beta_n^2)$$

Sum of squares of all the weight terms defined as β_0^2, β_1^2 etc.

→ In 2D space this can be viewed as weight matrix consisting of two terms say $\beta = [\beta_1, \beta_2]$

or we can say $\beta_i = \beta_1^2 + \beta_2^2$,

which defines a symmetrical circle around origin.

→ This symmetrical circle restricts the loss function from decreasing further than the value obtained on the intersection of the elliptical contour with the circle.

→ Since L_2 norm defines a circle as a boundary rather than weight values becoming 0, all the weight values shrink in proportion. Thus, though the model reduces the weight values, most of them still remain non-zero.

